GPU simulated quantum annealing for QUBO feature selection

The Quantum Annealers· NCHC / NVIDIA Open Hackathon · final presentation

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The Quantum Annealers

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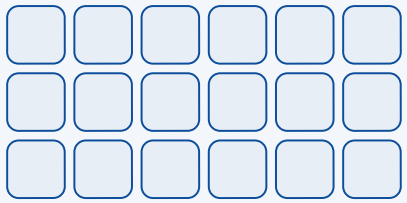
Team mentors



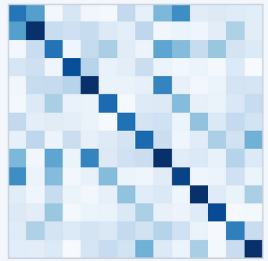
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The problem: Feature Selection - select best K features

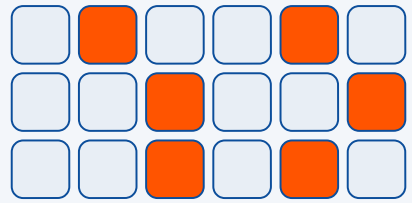


590,540 transactional data
431 features -> expand to
3000 features -> select **K**



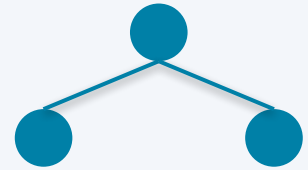
QUBO

Q_{ii} relevance
Q_{ij} redundancy
γ: exactly K



K = 500 chosen

from 3000 features.



fraud classifier
XGBoost, ROC-AUC

$$E(x) = -\alpha \sum_i r_i x_i + (1-\alpha) \sum_{i<j} d_{ij} x_i x_j + \gamma \left(\sum_i x_i - K \right)^2$$

C(3000, 500) ≈ 10^{585}

Impractical to brute force the optimal solution.

QUBO

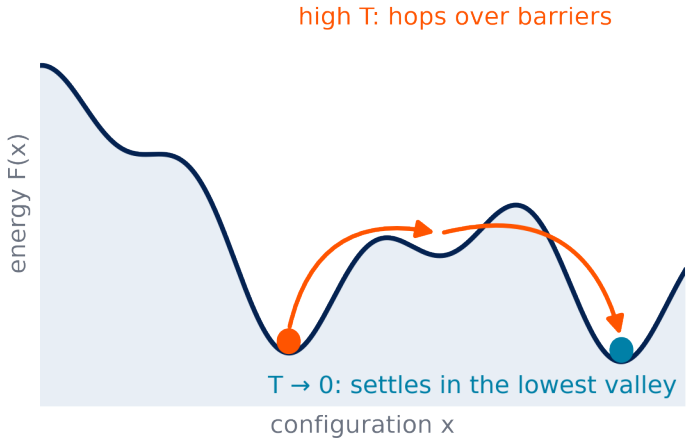
Quadratic Unconstrained Binary Optimization

low E(quality), fast (speed)

our job: a GPU solver that finds low-objective subsets fast

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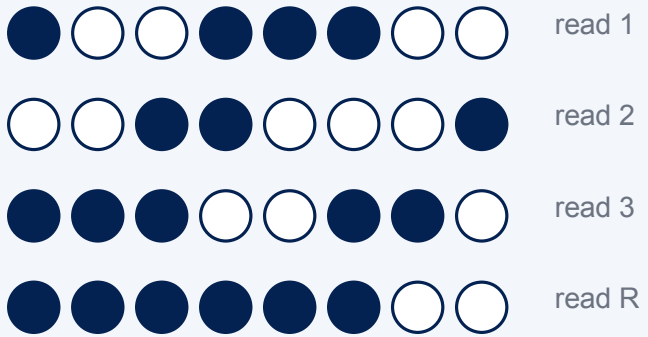
The motif: simulated annealing, and where it is parallel



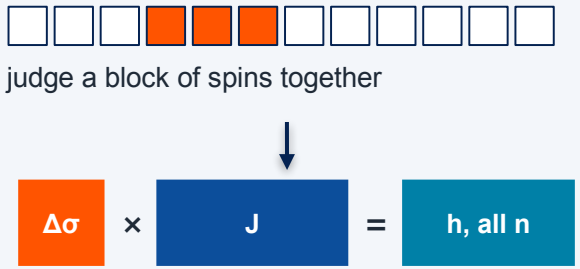
$$P(\text{flip}) = \min\left(1,\ e^{-\Delta E/T}\right)$$

2 parallelism opportunities:

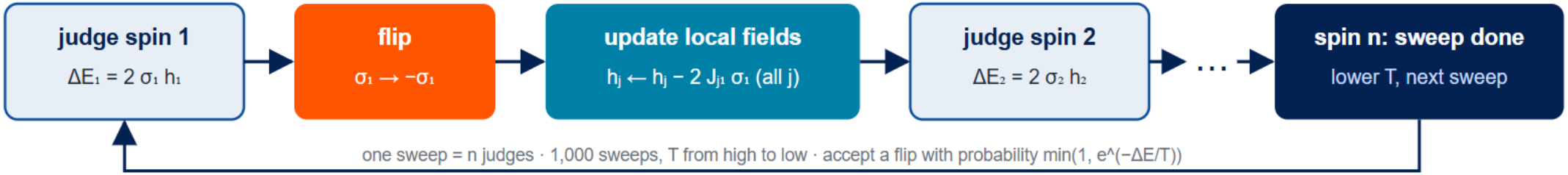
R independent chains (reads)



one sweep, 512 spins at a time



Monte Carlo Steps in plain:



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Next step: from classical to simulated quantum annealing

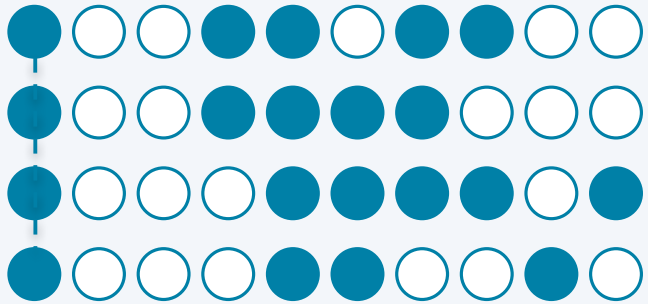
SA: one chain of n spins

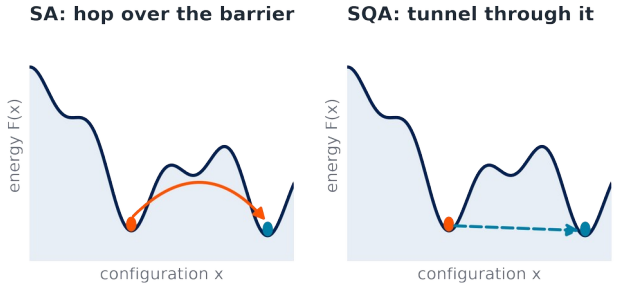


thermal hops over barriers; stuck once T is low

Suzuki–Trotter: M coupled copies

SQA: M coupled copies





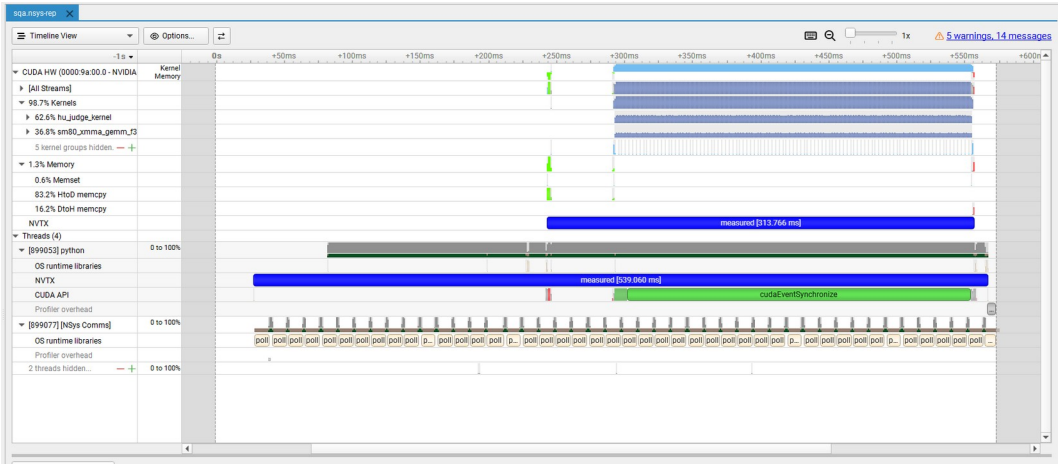
same spins, one more axis: the copies can tunnel through a barrier instead of climbing it

$$H = -\sum_{k=1}^{M}\left[\sum_{i<j}\frac{J_{ij}}{M}\,\sigma_{i,k}\sigma_{j,k} + \sum_{i}\frac{h_i}{M}\,\sigma_{i,k} + J_\perp \sum_{i}\sigma_{i,k}\,\sigma_{i,k+1}\right], \qquad J_\perp = -\frac{1}{2\beta}\ln\tanh\frac{\beta\Gamma}{M}$$

+ one more parallelism opportunity, parallel in M (trotters)

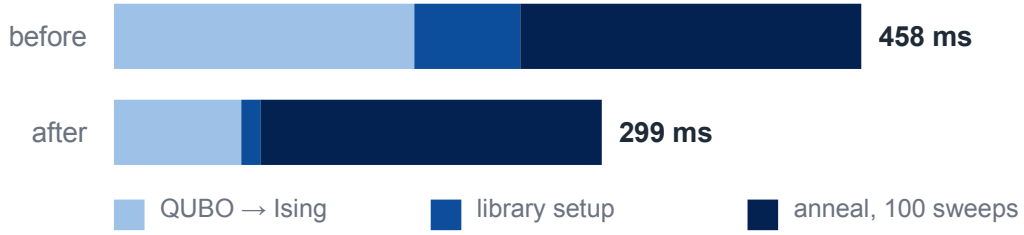
Reference paper: Y.-H. Chung, C.-J. Shih, S.-H. Hung, “Accelerating Simulated Quantum Annealing with GPU and Tensor Cores,” ISC High Performance 2022, LNCS 13289, Springer, 2022. doi:10.1007/978-3-031-07312-0_9

Profile first, then change: 2.60 → 1.41 ms per Monte Carlo step

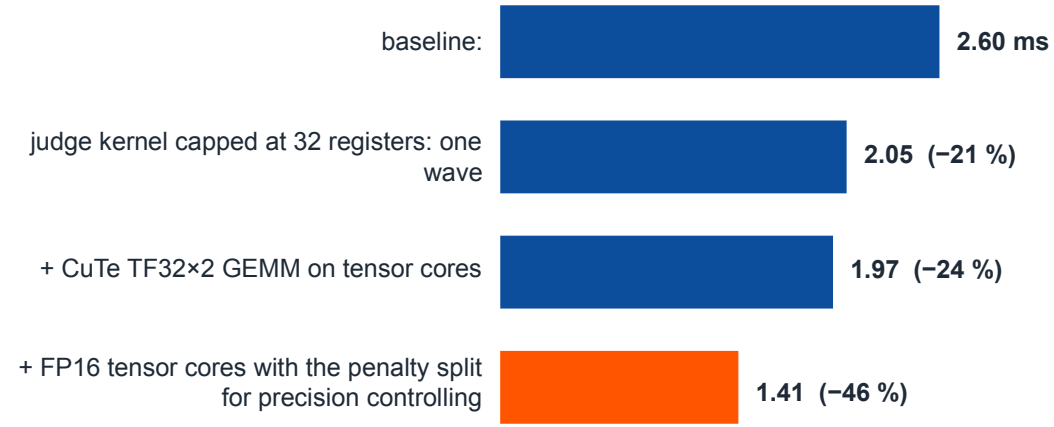


Nsight Systems, one D2 call: judge kernel 63 % · GEMM 37 % · GPU 98.5 % busy · judge kernel 1.21 waves, 44 registers · tensor cores 0 %

one 100-sweep D2 call: 458 → 299 ms



ms per MC step, D2 (M = 32, 50 reads, on H200)



1.84×

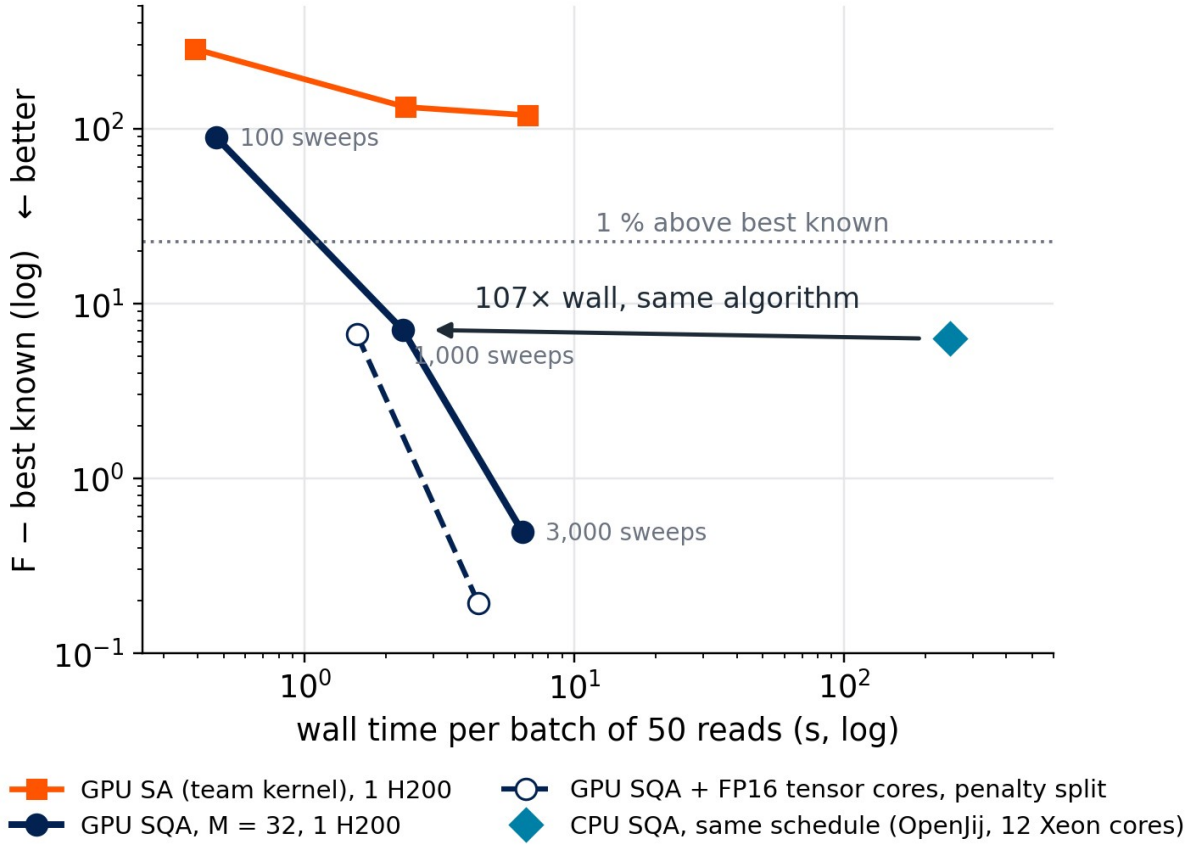
step time at equal quality; D2 3,000 sweeps 7.95 → 4.40 s

bit-exact

every kept change passed the --verify gate; the fast-but-wrong ones did not

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GPU-accelerated SQA: better answers, sooner



D2: n = 3,000, K = 500 · medians over 10 seeds · best known F = 2252.48 · H200 vs 12 Xeon cores

107×

wall time vs 12 Xeon cores, same algorithm and schedule: 2.3 s against 248 s (D2, 1,000 sweeps)

−119

objective points below GPU SA at 3,000 sweeps (2253 vs 2372);

0.58 vs 0.07

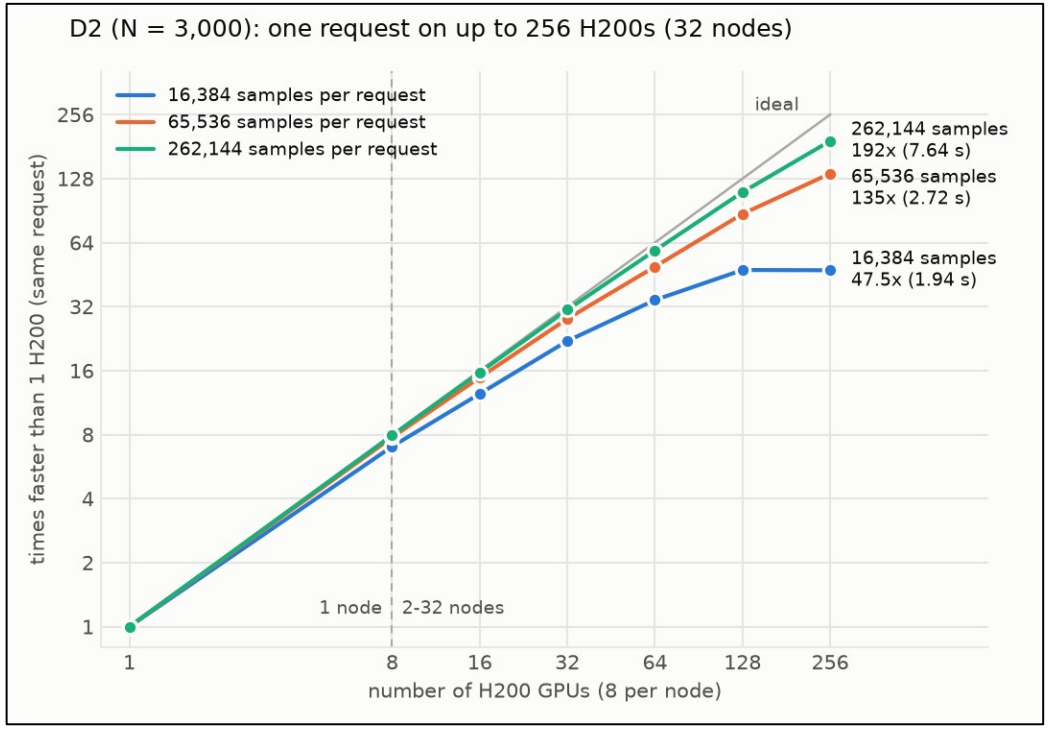
chance per read of hitting the exact optimum, SQA vs SA (brute-forced n = 32 sub-problems)

10 / 10

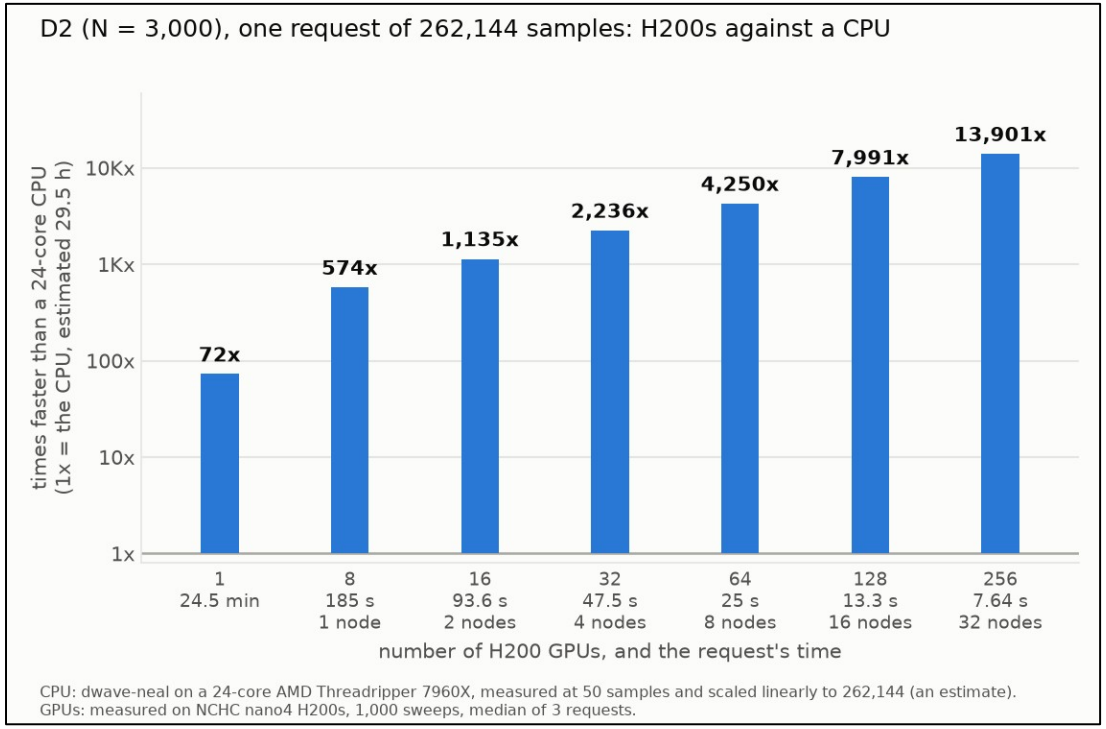
valid batches on D2.

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1 to 256 H200 GPUs: scaling comparison



Warp-SA, one request split over up to 32 nodes, similar scaling with SQA



Evolution and strategy: three weeks with AI agents

Week 1 · figuring out workflow

- No prior knowledge or GPU engineering.
- No GPU profiling experience.
- Haven't decide the workflow for the project.
- Little experience regarding working with AI agent.

Week 2 · testing workflow

- Using AI agent to create the draft code.
- Manually assess the codes logic.
- Using AI agents to speed up GPU engineering ideas brainstorming and implementation.

Week 3 · trustworthy workflow

- Provide the workflow framework below to AI agents.
- We only enter at how the assessment of the profiling and benchmarking lead to changes(including AI proposed changes).

the gate loop that made the agent trustworthy

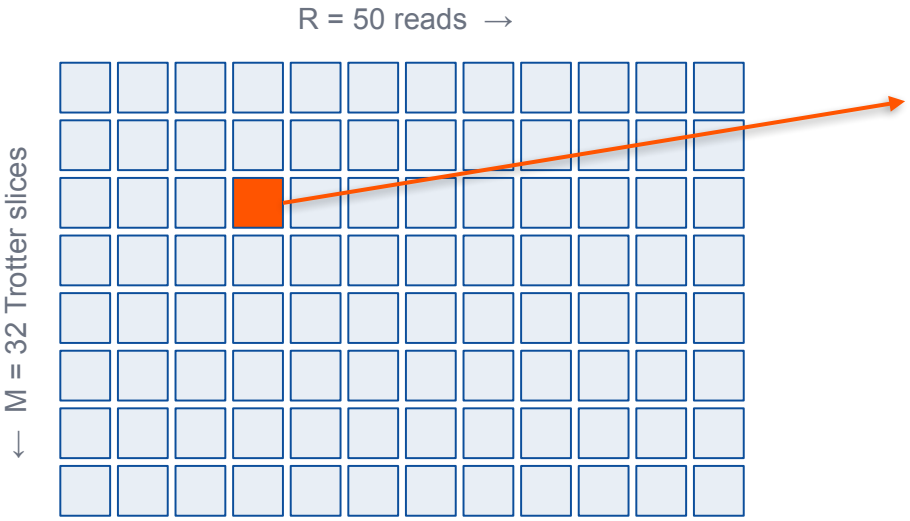


In conclusion

- QUBO optimization: Developed a solver for quadratic unconstrained binary optimization problems.
- Parallel simulated annealing: Exploited parallelism across independent runs (reads) and local-field updates.
- Simulated quantum annealing: Introduced an additional dimension of Trotter slices, enabling parallelism across the M slices.
- Computational performance: Achieved substantial speedups over the CPU implementation.
- AI-assisted development: Integrated AI agents into a structured workflow to accelerate research and engineering.
- Real-world application: Applied the pipeline to transaction fraud classification, demonstrating its practical applicability.


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GPU SQA: every slice of every read is one CUDA block



1,600 CUDA blocks on 132 SMs: every slice of every read sweeps at the same time

inside one block: one slice of one read



judge the 128 spins of the block in registers (Metropolis); their fields live in shared memory



one cuBLAS GEMM pushes the flips into every local field (the tensor-core half)



neighbouring slices judge alternate halves of each block, so the J⊥ coupling never races

48 + 24

judge launches + GEMMs per Monte Carlo step at n = 3,000 (24 blocks); 0.4 µs gaps, GPU 98.5 % busy

C = 1,600

columns in one matrix multiply: 32 slices × 50 reads, every one its own Markov chain

[reference]

[paper the implementation follows: author, year, venue]

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Problems: the bugs were cheap, the theory was not

Code problem

- Syntax bugs are easily captured by AI agents, so the implementation of the code is relatively fast.
- Probably estimated of 2-3 months of code implementation time needed if AI agents are not used, comparing to 1-2 weeks of code implementation done with AI agents.

Theory

- The theory of simulated quantum annealing is relatively complex than classical simulated annealing.
- To convert the ideas, the parallelism opportunities into real code implementation is a big barrier if without the help of AI agent to accelerate the process.
- In a physics background, with not much prior knowledge of GPU engineering, the implementation of the algorithm need a lot of the technical knowledge.

System

- Some nodes has some latency issue for sometimes, AI agent caught that.

Energy Efficiency

example from the template — replace with your screenshot

INPUTS	
# CPU Cores	64
# GPUs (A100)	6
Application Speedup	20.0x

Node Replacement	13.3x

GPU NODE POWER SAVINGS			
	AMD Dual Rome 7742	8x A100 80GB SXM4	Power Savings
Compute Power (W)	14,667	6,500	8,167
Networking Power (W)	619	93	526
Total Power (W)	**15,286**	**6,593**	8,693

Node Power efficiency	**2.3x**

ANNUAL ENERGY SAVINGS PER GPU NODE			
	AMD Dual Rome 7742	8x A100 80GB SXM4	Power Savings
Compute Power (kWh/year)	128,480	56,940	71,540
Networking Power (kWh/year)	5,424	814	4,610
Total Power (kWh/year)	**133,904**	**57,754**	**76,150**

$/kWh	$	0.18
Annual Cost Savings	$	13,707.04
3-year Cost Savings	$	41,121.13

Metric Tons of CO2	54
Gasoline Cars Driven for 1 year	12
Seedlings Trees grown for 10 years	892

(source: Link)

Calculator inputs for this project

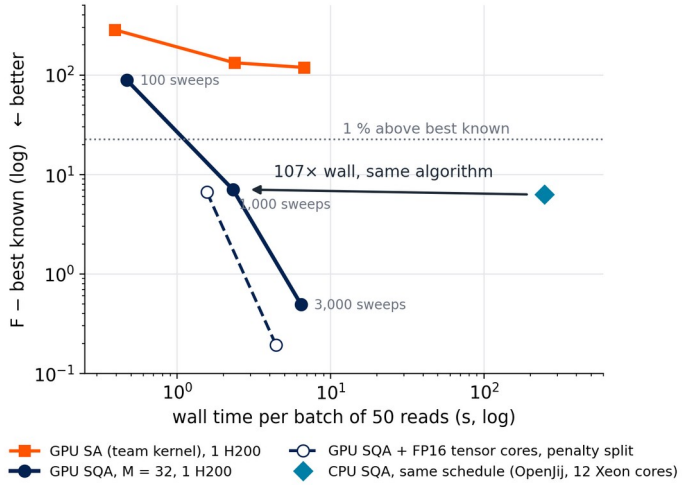
CPU cores: 12 (one GPU's share of a 2 × Xeon 8480+ node)
GPUs: 1 (H200)
application speed-up: 107× (D2, wall, same algorithm; D1rb: 20×)

≈ 80 CPU nodes replaced by one GPU node (D2)

calculator: Google Sheet from the template

Application Background

Feature selection for fraud detection as a QUBO: choose K of n transaction features (n = 3,000, K = 500) that minimise $F = x^TQx + \gamma(\Sigma x - K)^2$. Motif: simulated (quantum) annealing — Monte-Carlo sweeps over spins, parallel over independent reads and Trotter slices, local fields by matrix multiply.



one H200 reaches in 2.3 s what 12 CPU cores reach in 248 s (D2)

Hackathon Objectives and Approach

Programming models: CUDA C++ with cuBLAS and CUTLASS CuTe; NVIDIA Warp (Python) for Warp-SA.
Profiling: Nsight Systems / Compute — judge kernel 63 %, GEMM 37 %, 1.21 waves, tensor cores idle.
Tuning: register cap (one wave), FP16 tensor cores with an exact penalty split, host setup removed.
Multi-GPU: reads across 1–256 H200s, no inter-GPU communication.

Technical Accomplishments and Impact

One H200 = 107× twelve Xeon cores at equal quality; SQA reaches a 5 % lower objective than SA with every batch valid; step time 2.60 → 1.41 ms; 99.6 % weak-scaling efficiency on 256 GPUs; Warp-SA 192× on 256 GPUs.
A minutes-long CPU step becomes seconds — and the downstream study showed the formulation, not the solver, is the next lever.

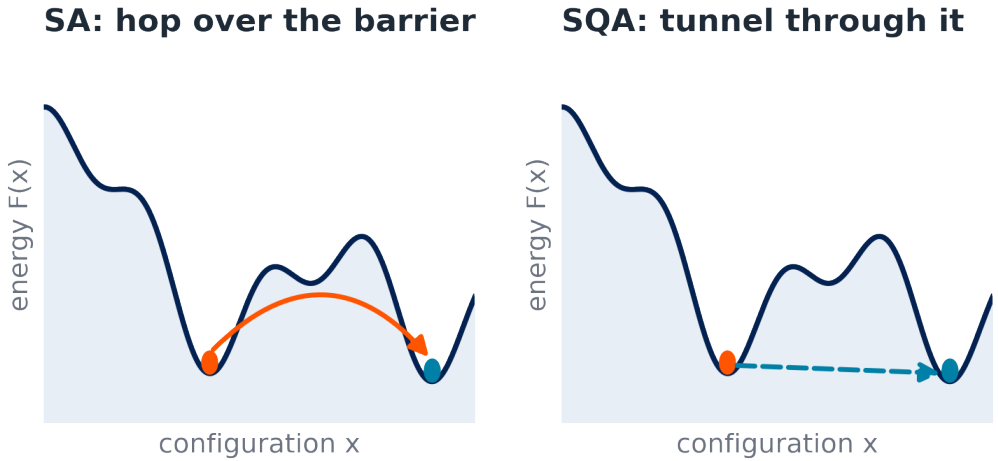
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GPU 模擬量子退火：秒級完成「三千選五百」的特徵選取

[__] 團隊來自臺灣大學 [__] 老師帶領的 [__] 實驗室，
AI Agent 將 GPU 程式開發與效能測試流程加速了 [__] 倍；
GPU 將 QUBO 特徵選取的求解加速了 107 倍（單張 H200 對 12 核 CPU），256 張 H200 再快 192 倍!

信用卡詐欺偵測模型要從數千個交易特徵中挑出最有用的一組：從 3,000 個特徵挑 500 個，組合數超過 10 的 585 次方，無法窮舉。我們把它寫成 QUBO（二次無約束二元最佳化）問題，先用模擬退火求解；再引入模擬量子退火（SQA），用 M 個相互耦合的古典副本模擬量子穿隧，解的品質比模擬退火好 5 %，而且每一批都滿足「剛好選 K 個」的限制。
在 GPU 上，每個副本、每次讀取各佔一個 CUDA 區塊，局部場更新交給一次矩陣乘法；用 Nsight 找出瓶頸後，把 judge kernel 壓進一個 wave，並用 Tensor Core 在不損失精度的前提下再快 1.47 倍（每步 2.60 → 1.41 ms）。單張 H200 比 12 核 CPU 快 107 倍；讀取數分散到 256 張 H200 時，弱擴展效率 99.6 %。研究上的發現：目標函數更低並未讓分類器更好，下一步的槓杆在問題的建模而不是求解器。

D2 （n = 3,000, K = 500）： SQA 目標值 2253 vs SA 2372 （3,000 sweeps）；一張 H200 2.3 s vs OpenJij 12 核 248 s （107 倍）
精確解命中率（n = 32）： SQA 0.58 vs SA 0.07；每步時間 2.60 → 1.41 ms （等品質）
256 張 H200：每秒 7.4 × 10^{11} 次自旋更新，效率 99.6 %； Warp-SA 262,144 個樣本 192 倍



示意圖：模擬退火翻越能障，模擬量子退火穿隧而過

